

Ge Yan

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PhD candidate, Electrical Engineering, National University of Singapore, Singapore

PROFESSIONAL SUMMARY

I am a wireless researcher specializing in communication and sensing systems, with expertise in mathematical modeling, optimization, and computational electromagnetics. I develop and implement algorithms for reconfigurable MIMO, channel knowledge maps, radio sensing, and inverse problems, with additional experience applying machine learning to wireless and sensing applications.

TECHNICAL EXPERTISE

- **Wireless:** MIMO, beamforming, channel models, computational electromagnetics.
- **Algorithms:** optimization, numerical methods, signal processing, inverse problems.
- **Machine learning:** deep learning, computer vision, PyTorch, TensorFlow.
- **Programming & tools:** Python, MATLAB, C/C++, COMSOL, CST, CVX, Git, Linux.

EDUCATION

- **2022/08-2026.10 (expected)** **PhD. in Electrical and Computer Engineering**
 - Institution: National University of Singapore, Singapore
 - Advisors: Prof. Rui Zhang, Prof. Xudong Chen
 - Thesis Title: Reconfigurable MIMO for Wireless Communication: Optimization and Practical Channel Knowledge Acquisition
- **2021/01-2022/06** **MSc. in Electrical and Computer Engineering**
 - Institution: National University of Singapore, Singapore
 - GPA: 4.75/5.00
- **2016/09-2020/06** **B.Eng. in Information Science & Engineering**
 - Institution: Chien-Shiung Wu College, Southeast University, Nanjing, China
 - GPA: 3.89/4.00 (Overall Ranking: 1/114)

INDUSTRY EXPERIENCE

- **2020/08-2020/12** **AI Research Intern, R&D, SenseTime, Shanghai, China**
 - Participated in image recognition and segmentation projects.
 - Investigated GAN-based video generation.

RESEARCH EXPERIENCE

- **[Ongoing] Channel Knowledge Map (CKM) Modelling from An Electromagnetic Perspective (2026/04-present)**
 - Developing an electromagnetic approach to CKM modeling and reconstruction in contrast to conventional ray-tracing-based approaches.
 - Represent wireless channels in a physically consistent manner without requiring detailed geometric descriptions of the environment.

- **[Ongoing] Slow Movable Antenna (MA)-Aided Wireless Communications Based on Long-Term Channel Statistics (2024/04-present)**
 - Proposed a novel design framework for MA-aided wireless communications based on long-term channel statistics instead of instantaneous channel.
 - Alleviated the burden of frequent channel acquisition and antenna repositioning, thereby facilitating practical deployment.
- **Practical Configuration for Intelligent Reflecting Surface (IRS)-Assisted Wireless Systems (2022/10-2025/09)**
 - Developed practical channel acquisition and quasi-static optimization methods for IRS-aided wireless systems, enabling cost-efficient long-term coverage enhancement.
 - Utilized channel measurements available from existing protocols without additional signalling, facilitating seamless integration of IRSs into existing networks.

SELECTED PROJECTS & PROTOTYPING

- **Movable Antenna (MA)-Aided Wireless Communications Prototype (2022/10-2025/09)**
 - Developed a prototype with mechanical antenna positioning controls and integrated it with National Instruments software-defined radio platforms.
 - Designed and conducted experiments to validate the performance of MA-aided wireless communication algorithms under practical system conditions.
 - Led algorithm integration, system debugging, and performance evaluation.
- **Intelligent Reflecting Surface (IRS) Channel Acquisition Prototype (2022/10-2025/09)**
 - Collaborated with a remote implementation team to integrate algorithms from my research into a prototype of IRS-aided wireless communication systems.
 - Provided algorithm implementation and technical support, assisted with system debugging, and contributed to performance evaluation and validation.

SELECTED PUBLICATIONS

1. **G. Yan**, L. Zhu, W. Ma, and R. Zhang, "Slow Movable Antenna System Design Based on Cell-Specific Long-Term Angular Power Spectrum," 2026. [Online]. Available: <https://arxiv.org/abs/2605.12192>. (submitted to *IEEE Trans. Wireless Commun.*)
2. **G. Yan**, L. Zhu, and R. Zhang, "Movable Antenna Aided Multiuser Communications: Antenna Position Optimization Based on Statistical Channel Information," in *IEEE Trans. Commun.*, vol. 74, pp. 1793-1810, 2026.
3. **G. Yan**, L. Zhu, and R. Zhang, "Power Measurement Enabled Channel Autocorrelation Matrix Estimation for IRS-Assisted Wireless Communication," in *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 1832-1848, Mar. 2025.
4. **G. Yan**, L. Zhu, H. Sun, and R. Zhang, "Wideband Coverage Enhancement for IRS-Aided Wireless Networks Based on Power Measurement," in *IEEE Wireless Commun. Lett.*, vol. 15, pp. 670-674, 2026.

AWARDS/FELLOWSHIPS

- President's Graduate Fellowship (2022/08-2026/07) National University of Singapore
- National Scholarship (Top 1%, 2018/10) Chinese Ministry of Education
- President Scholarship (Top 1%, 2017/10) Southeast University, Nanjing, China